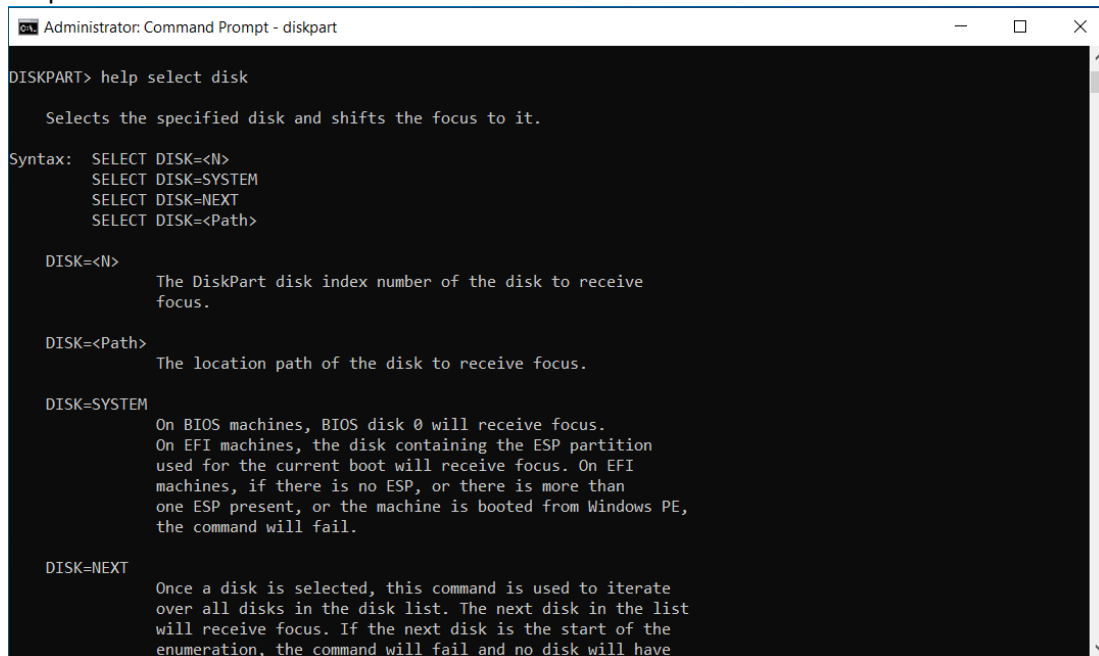


Activity 4-1: Using DiskPart

1. Opened a Command Prompt (as administrator), started Diskpart, and learned how to use the Help feature.



```
Administrator: Command Prompt - diskpart

DISKPART> help select disk

    Selects the specified disk and shifts the focus to it.

Syntax:  SELECT DISK=<N>
         SELECT DISK=SYSTEM
         SELECT DISK=NEXT
         SELECT DISK=<Path>

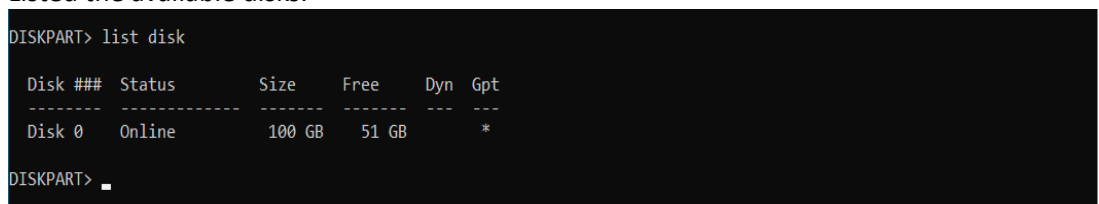
DISK=<N>
    The DiskPart disk index number of the disk to receive
    focus.

DISK=<Path>
    The location path of the disk to receive focus.

DISK=SYSTEM
    On BIOS machines, BIOS disk 0 will receive focus.
    On EFI machines, the disk containing the ESP partition
    used for the current boot will receive focus. On EFI
    machines, if there is no ESP, or there is more than
    one ESP present, or the machine is booted from Windows PE,
    the command will fail.

DISK=NEXT
    Once a disk is selected, this command is used to iterate
    over all disks in the disk list. The next disk in the list
    will receive focus. If the next disk is the start of the
    enumeration, the command will fail and no disk will have
```

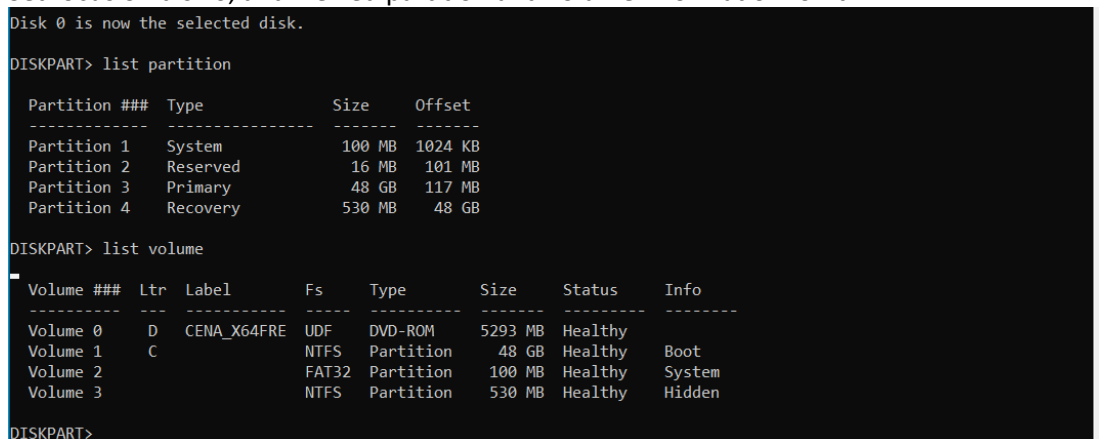
2. Listed the available disks.



```
DISKPART> list disk

Disk ###  Status       Size      Free      Dyn  Gpt
-----  -
Disk 0     Online       100 GB    51 GB          *
```

3. Set focus on disk 0, and viewed partition and volume information for it.



```
Disk 0 is now the selected disk.

DISKPART> list partition

Partition ###  Type              Size      Offset
-----  -
Partition 1    System           100 MB    1024 KB
Partition 2    Reserved         16 MB     101 MB
Partition 3    Primary          48 GB     117 MB
Partition 4    Recovery         530 MB     48 GB

DISKPART> list volume

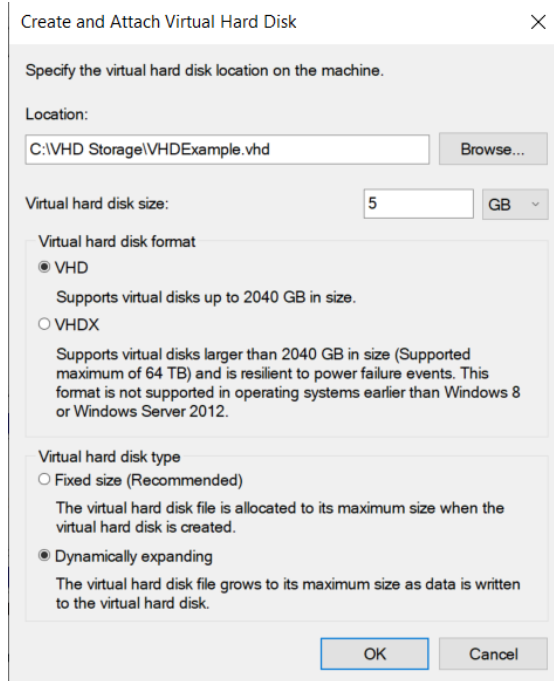
Volume ###  Ltr  Label        Fs      Type          Size      Status      Info
-----  -
Volume 0     D    CENA_X64FRE  UDF     DVD-ROM       5293 MB   Healthy
Volume 1     C    NTFS         NTFS     Partition     48 GB    Healthy    Boot
Volume 2     FAT32  Partition    100 MB   Healthy    System
Volume 3     NTFS   Partition    530 MB   Healthy    Hidden

DISKPART>
```

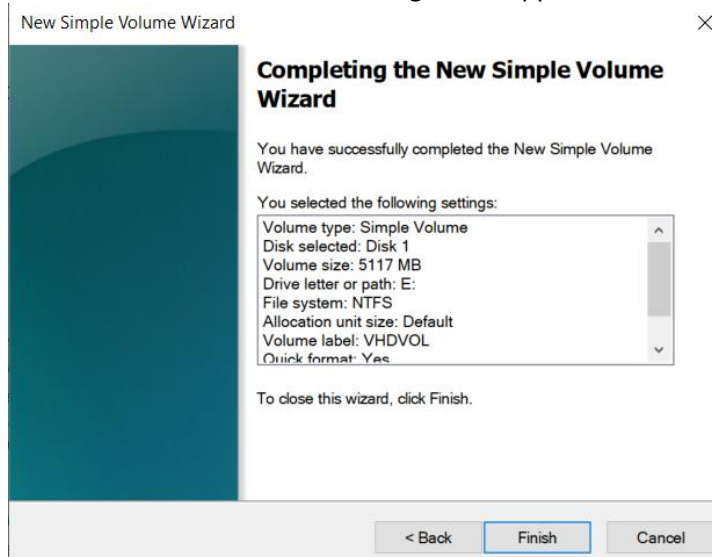
4. Exited Disk part and closed the Command Prompt window.

Activity 4-2: Creating VHD Disks

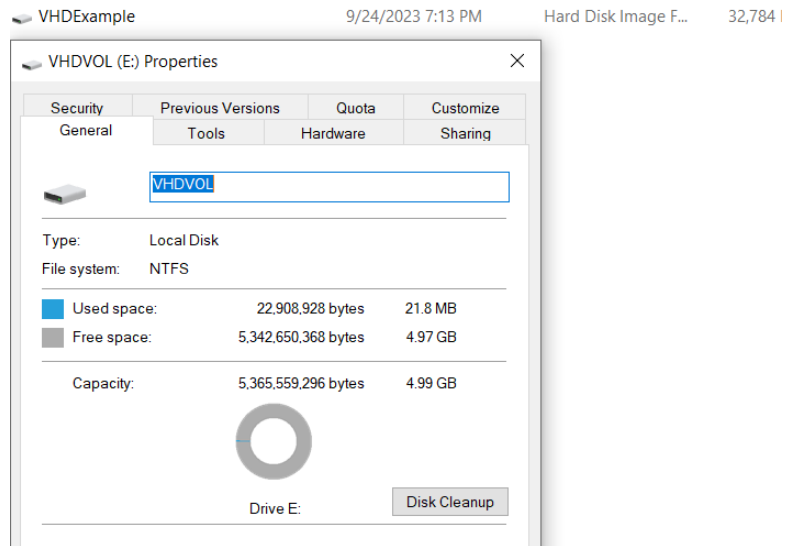
1. Created a C:\VHD Storage Folder, and opened Disk Management.
2. Created VHD VHDExample with the settings shown below:



3. Viewed the VHD in the Disk Management app. Initialized the disk, and created VHDVOL.



4. Examined the actual VHD size in the C:\VHD Storage folder. While it was configured as 5 GB, the actual storage used at this time was much less.



Activity 4-3: Managing VHD Disks

1. Opened a Command Prompt as Administrator, and selected, viewed, and dismounted the virtual disk.

```
Administrator: Command Prompt - diskpart
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>diskpart

Microsoft DiskPart version 10.0.19041.964

Copyright (C) Microsoft Corporation.
On computer: MOS-WIN10-HSK

DISKPART> select vdisk file="C:\VHD Storage\VHDEExample.vhd"

DiskPart successfully selected the virtual disk file.

DISKPART> detail vdisk

Device type ID: 2 (VHD)
Vendor ID: {EC984AEC-A0F9-47E9-901F-71415A66345B} (Microsoft Corporation)
State: Attached not open
Virtual size: 5120 MB
Physical size: 32 MB
Filename: C:\VHD Storage\VHDEExample.vhd
Is Child: No
Parent Filename:
Associated disk#: 1

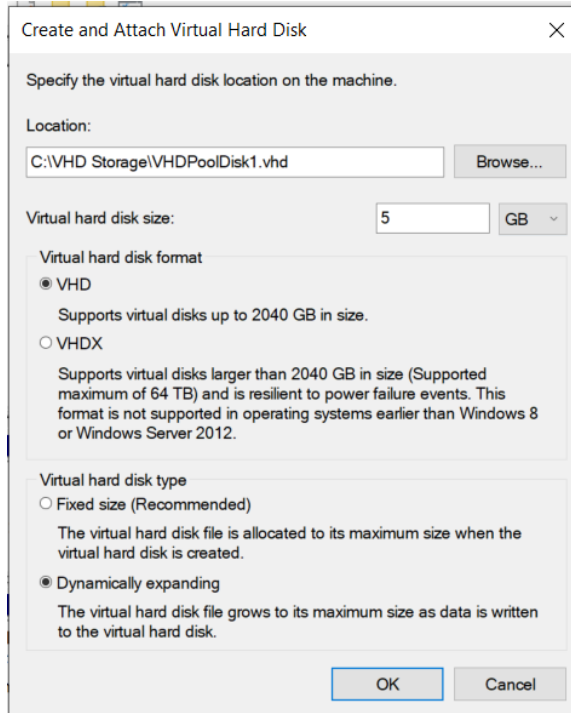
DISKPART> detach vdisk

DiskPart successfully detached the virtual disk file.

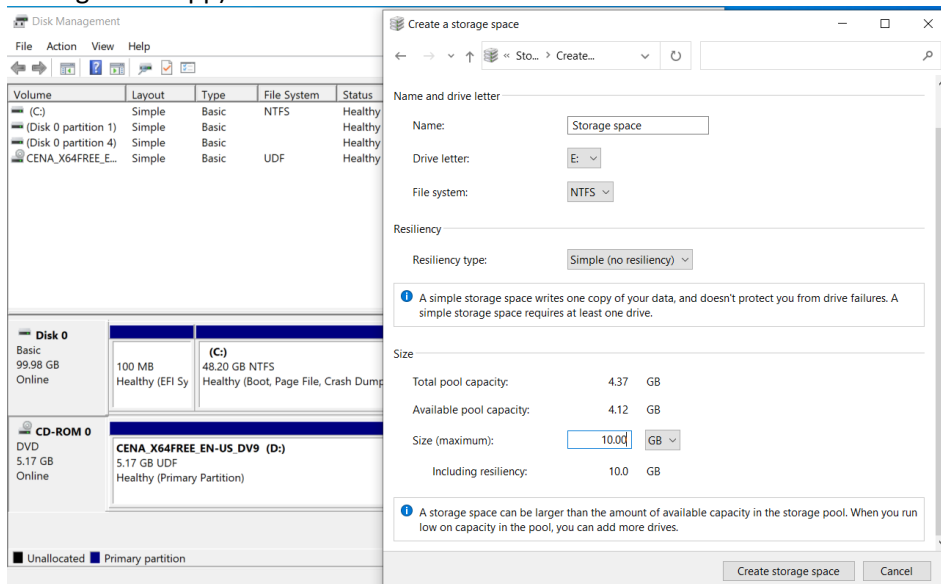
DISKPART>
```

Activity 4-4: Creating the Initial Storage Pool and Storage Space Volumes

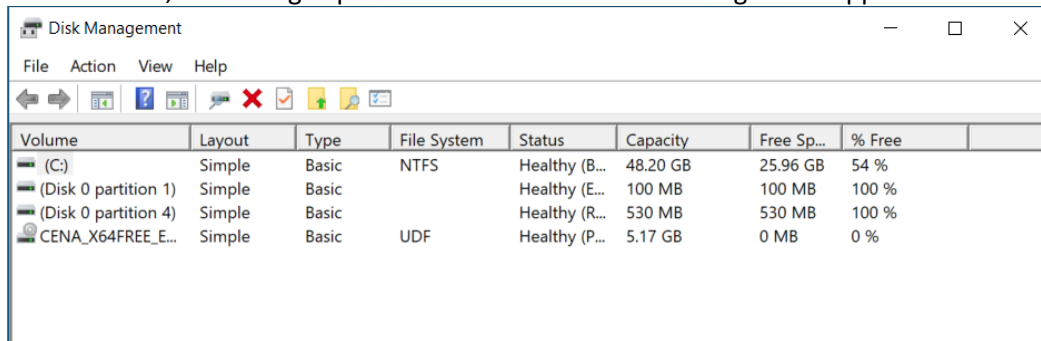
1. Created the VHDPoolDisk1.vhd with the settings shown below.



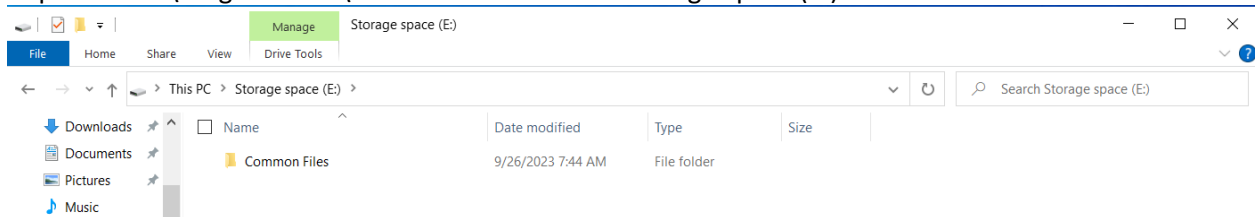
2. Created the new Storage Space. Noted that it disappeared from the Disk Management when the vhd was assigned to the storage pool. Settings for the Storage space are shown below. (I closed the Disk management mistakenly and when I open, Storage Space was visible in the Disk Management app)



- Once created, the Storage Space was visible in the Disk Management app.

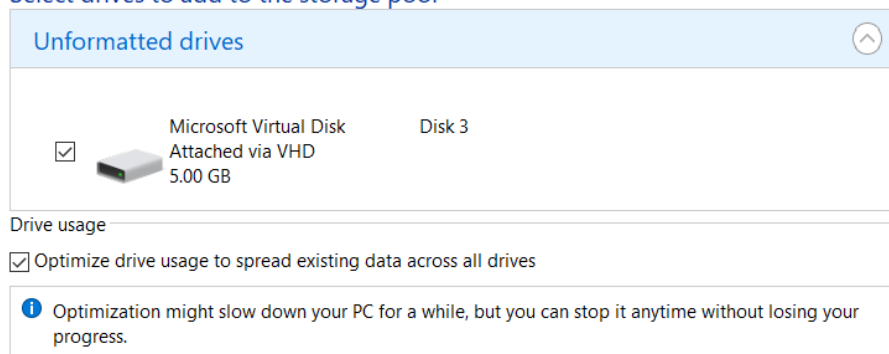


- Copied the C:\Program Files\Common Files into the Storage Space (E:)

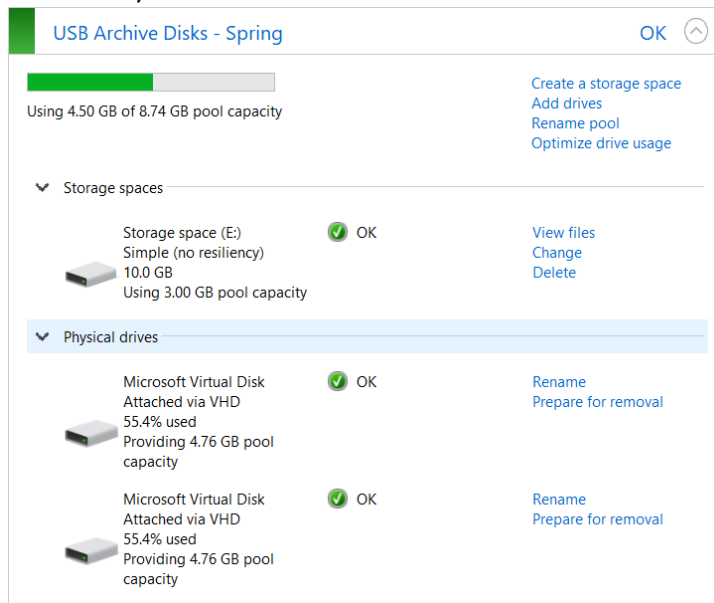


Activity 4-5: Managing Storage Pools and Storage Space Volumes

- Used Disk Management to create VHDPoolDisk2.
- Opened Storage Space Management, elevated privileges, and added a drive to the pool.
[Select drives to add to the storage pool](#)

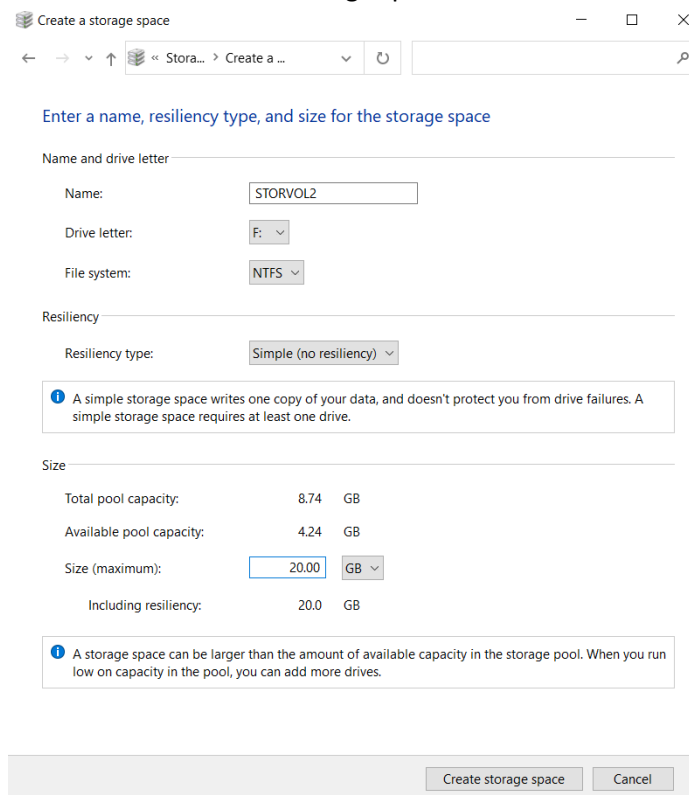


3. Viewed Physical Disks associated with the Pool.

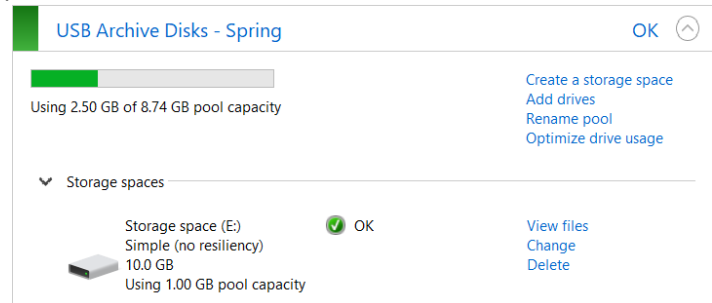
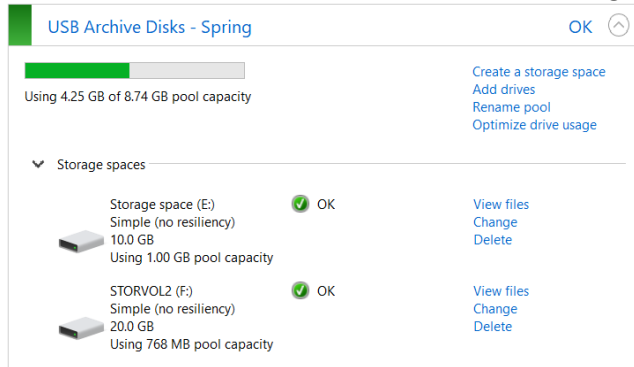


4. Reviewed the “Prepare for Removal” option and cancelled the process.

5. Created the STORVOL2 storage space.



6. Viewed, and then Deleted the STORVOL2 storage space.



Activity 4-6: Creating Fault-Tolerant Storage Space

1. Creating STORVOL3, it cannot be created with Parity, because there are not enough disks in the pool. The create option is greyed-out.

[Enter a name, resiliency type, and size for the storage space](#)

Name and drive letter

Name:

Drive letter:

File system:

Resiliency

Resiliency type:

i A parity storage space writes your data with parity information, helping to protect you from a single drive failure. A parity storage space requires at least three drives.

Size

Total pool capacity: 8.74 GB

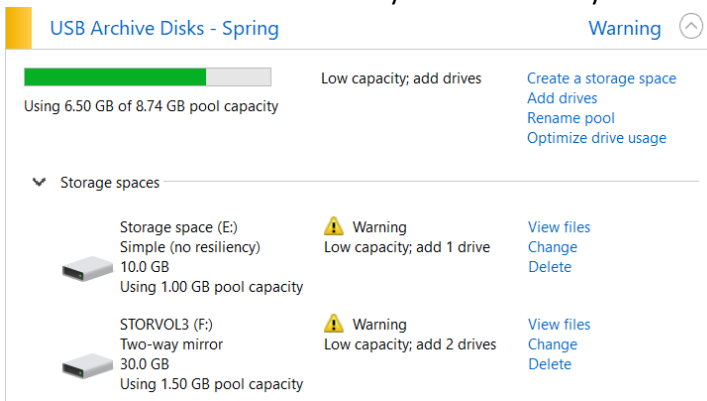
Available pool capacity: 6.24 GB

Size (maximum): GB

Including resiliency: 45.0 GB

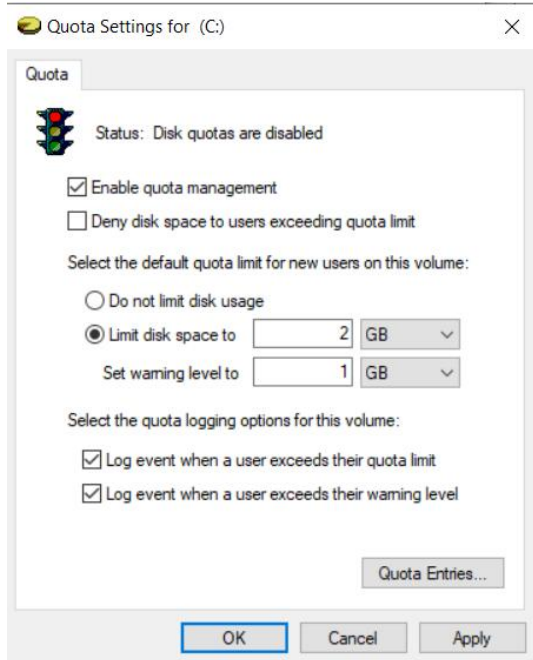
i A storage space can be larger than the amount of available capacity in the storage pool. When you run low on capacity in the pool, you can add more drives.

2. Created STORVOL3 with Two-way mirror resiliency.



Activity 4-7: Enabling Disk Quotas for NTFS Partitions

1. Enabled quota management on the C: drive.



2. Waited for the Disk Quota status to change to Active, and then checked quota entries.

Quota Entries for (C:)

Status	Na...	Logon Name	Amount Used	Quota Limit	Warning Level	Percent Used
!	Abo...	NT SERVICE...	8.99 GB	2 GB	1 GB	449
!	Abo...	NT AUTHO...	8.34 GB	2 GB	1 GB	417
!	War...	MOS-WIN1...	1.01 GB	2 GB	1 GB	50
!	War...	NT AUTHO...	1.66 GB	2 GB	1 GB	83
+	OK	BUILTIN\Ad...	7.51 GB	No Limit	No Limit	N/A
+	OK	NT AUTHO...	152.09 MB	2 GB	1 GB	7
+	OK	NT SERVICE...	1 KB	2 GB	1 GB	0
+	OK	[Ac... S-1-5-80-26...	1 KB	2 GB	1 GB	0
+	OK	[Ac... S-1-5-21-39...	1 MB	2 GB	1 GB	0
+	OK	[Ac... S-1-5-21-24...	2 KB	2 GB	1 GB	0

- Opened a command prompt in Administrative mode and issued the “fsutil quota query C:” command.

The screenshot displays two windows side-by-side. The left window is titled "Quota Entries for (C:)" and shows a table of quota information. The right window is titled "Administrator: Command Prompt" and shows the output of the command "fsutil quota query C:".

Status	Na...	Logon Name	Amount Used	Quota Limit	Warning Level	Percent Used
Abo...	NT SERVICE...	8.99 GB	2 GB	1 GB	449	
Abo...	NT AUTHO...	8.33 GB	2 GB	1 GB	416	
War...	MOS-WIN1...	1.01 GB	2 GB	1 GB	50	
War...	NT AUTHO...	1.66 GB	2 GB	1 GB	83	
OK	BUILTIN\Ad...	7.51 GB	No Limit	No Limit	N/A	
OK	NT AUTHO...	152.22 MB	2 GB	1 GB	7	
OK	NT SERVICE...	1 KB	2 GB	1 GB	0	
OK	[Ac... S-1-5-80-26...	1 KB	2 GB	1 GB	0	
OK	[Ac... S-1-5-21-39...	1 MB	2 GB	1 GB	0	
OK	[Ac... S-1-5-21-24...	2 KB	2 GB	1 GB	0	

```

Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.3448]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>fsutil quota query C:
FileSystemControlFlags = 0x00000031
Quotas are tracked on this volume
Logging enable for quota limits and threshold
The quota values are up to date

Default Quota Threshold = 0x0000000040000000
Default Quota Limit = 0x0000000080000000

SID Name = BUILTIN\Administrators (Alias)
Change time = Thursday, August 31, 2023 19:36:51 PM
Quota Used = 8072334336
Quota Threshold = 18446744073709551615
Quota Limit = 18446744073709551615

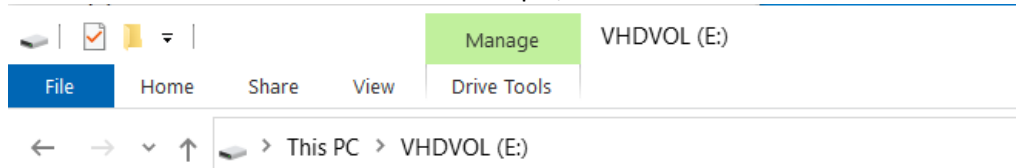
SID Name = NT SERVICE\TrustedInstaller (WellKnownGroup)
Change time = Tuesday, September 26, 2023 19:22:16 PM
Quota Used = 9659732992
Quota Threshold = 1073741824
Quota Limit = 2147483648

SID Name = MOS-WIN10-HSK\User1 (User)
Change time = Tuesday, September 26, 2023 19:27:24 PM
Quota Used = 1100260352
Quota Threshold = 1073741824
Quota Limit = 2147483648
  
```

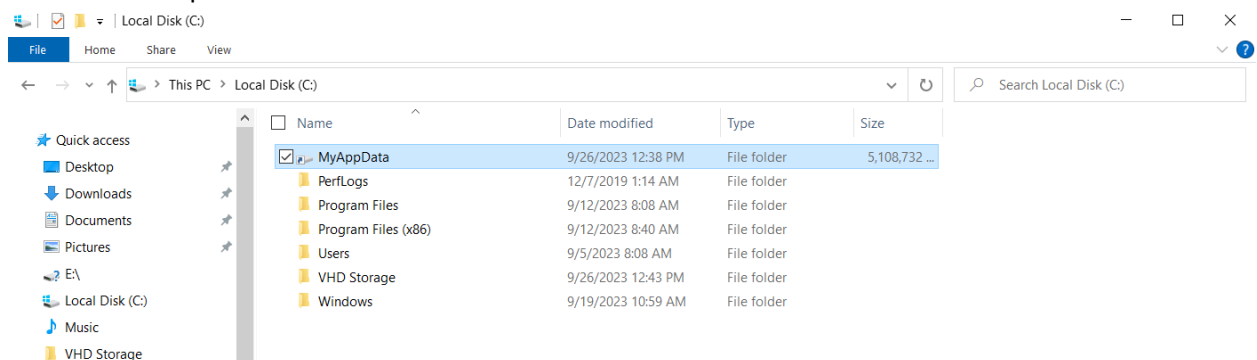
- Closed all open windows.

Activity 4-8: Managing Mount Points

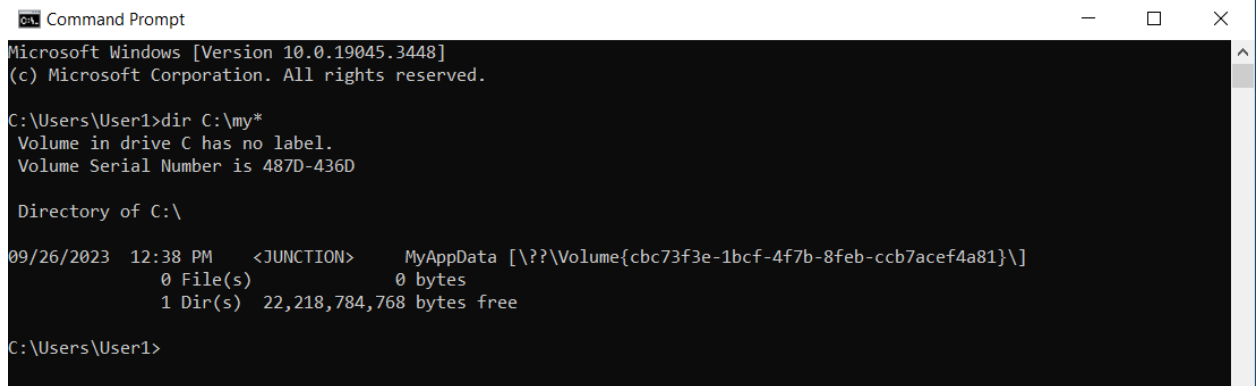
- Created C:\MyAppData folder.
- Browsed to and double-clicked on VHDExample, which mounted the vhd as volume E:



- Created DataFile, and closed Windows Explorer
- Removed the drive letter (E:) from the VHDVOL, and added the path of the C:\MyAppData as the volume mount point.



5. Viewed the folder at the command prompt. Noted that it is shown as a “Junction”.



```
Command Prompt
Microsoft Windows [Version 10.0.19045.3448]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User1>dir C:\my*
Volume in drive C has no label.
Volume Serial Number is 487D-436D

Directory of C:\

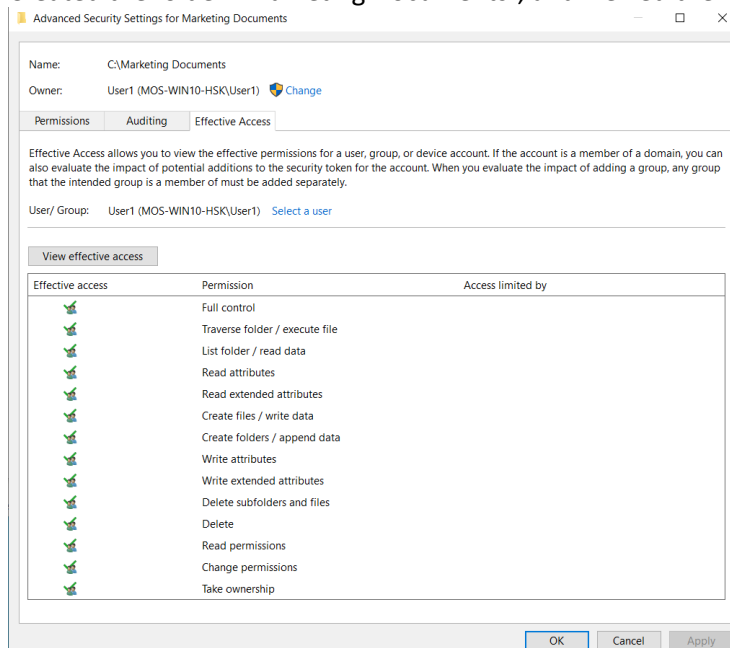
09/26/2023  12:38 PM    <JUNCTION>        MyAppData [\\?\Volume{cbc73f3e-1bcf-4f7b-8feb-ccb7acef4a81}\]
               0 File(s)                0 bytes
               1 Dir(s)  22,218,784,768 bytes free

C:\Users\User1>
```

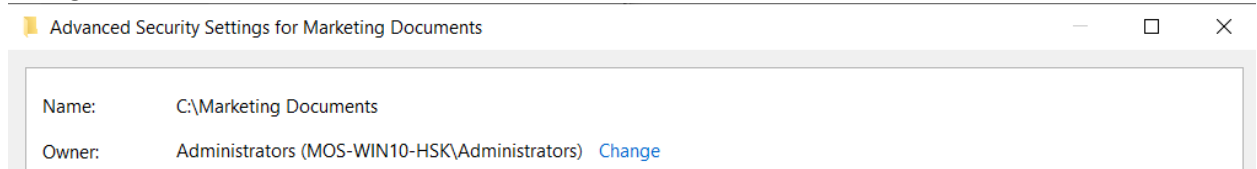
6. Detached the VHD, and closed disk management.

Activity 4-9: Managing File and Folder Permissions

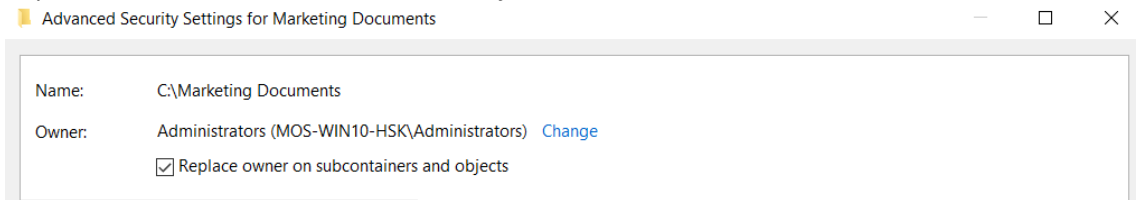
1. Created the folder “Marketing Documents”, and viewed the Effective Access for User1.



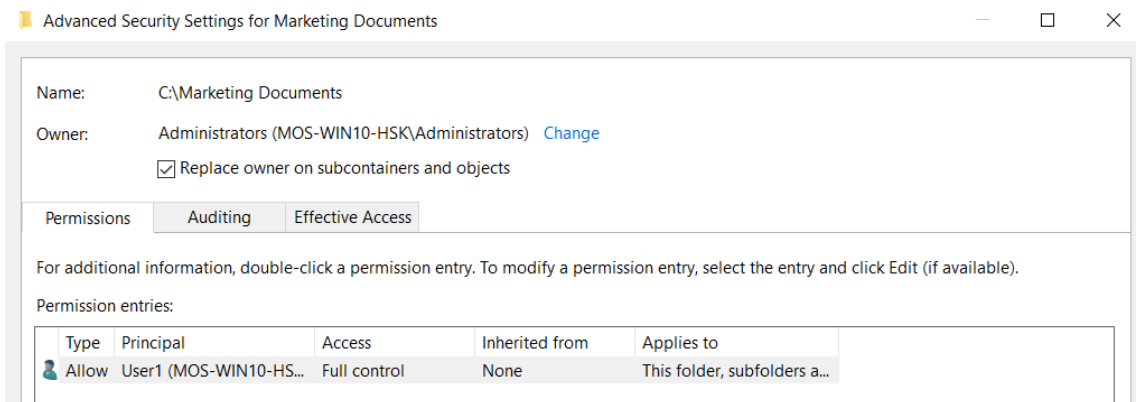
2. Changed the Owner to Administrators.



3. Replaced owner of sub-containers and objects.



4. Clicked the “Disable Inheritance” button, and selected the option to “Remove all...”
5. Added User00 as a security principal, noted that the new security entry “Inherited From” column indicates “None” (the permissions are granted explicitly at this level). (There is no warning display during the process)



6. Closed and re-opened the Properties of the folder, and clicked on the Security tab, observed that now, only User1 was listed in the Access Control Entries. (Left below)
7. Created the “First Quarter Report.txt” document and observed the permissions that it inherited from the folder’s settings. (Right below).

